

Experiments with a Single Factor: ANOVA



Today's lecture

- We will talk about
 - One-factor experiments
 - ANOVA
 - Reading material: section 3.1-3.3

Background Information

- In last lecture, we reviewed **statistical methods** for **comparing two treatments** (two conditions)
 - For example, use two-sample t-test to compare two-population means
- **Another way** to describe this experiment is a **single-factor experiment** with **two levels of the factor**
 - The levels are the two different populations
 - **For example**: we might want to compare the average height of a female group vs a male group at a chosen school.
 - The factor is gender
 - The two levels are male and female.

Background Information – cont'd

- Many experiments of this type involve **more than two levels** of the factor.
- This lecture, we will focus on **methods** for the **design** and **analysis** of **single-factor experiment** with **arbitrary number** of **levels** of the factor
 - For example, a levels (or a treatments)
 - We will assume that the experiment has been **completely randomized**

Example #1: single-factor Experiment

- Response: starting salary of an RIT graduate
- Factor: college within RIT that the students attended
 - We have 8 colleges (COS, COE, etc..)
 - Number of levels of the factor: 8
- **Goal** of this study: to **compare** the average starting salaries ($\mu_1, \mu_2, \mu_3, \mu_4, \mu_5, \mu_6, \mu_7, \mu_8$)
- **Recall**
 - Compare one college average mean μ_1 to some pre-specified value, one sample t-test
 - Compare two college average means $\mu_1 = \mu_1$??, two sample t-test
- what if **≥ 3** ?

Example #1 – cont'd

- ≥ 3
- For example: $H_0: \mu_1 = \mu_2 = \mu_3$
- Statistical test?
 - Could do numerous two sample t-test: $H_0: \mu_1 = \mu_2$, $H_0: \mu_2 = \mu_3$, $H_0: \mu_1 = \mu_3$???
 - This is not the best solution to this problem
 - This approach does not test equality of all three pairs above simultaneously
 - Performing all pairwise t-test is inefficient
 - Conducting all those pair comparison inflates the type I error
Why??? -> see explanations on next slide

Conducting all those pair comparison inflates the type I error -- WHY?

- Suppose we test all **three means** are equal
 - do numerous two sample t-test: $H_0: \mu_1 = \mu_2$, $H_0: \mu_2 = \mu_3$, $H_0: \mu_1 = \mu_3$
- If we select **$\alpha = 0.05$**
- The probability of reaching the correct decision on any **single comparison** is $1 - 0.05 = 0.95$
- Three tests $H_0: \mu_1 = \mu_2$, $H_0: \mu_2 = \mu_3$, $H_0: \mu_1 = \mu_3$ are conducted
- The probability of reaching the correct decision on **all three comparison** is $(1 - 0.05)^3 = 0.95^3$ (less than 0.95)
- The overall **probability of type I error** for the set of tests
 - $= 1 - (1 - 0.05)^3 = 1 - 0.95^3 \approx 0.1426$ (> 0.05)
 - So the type I error is **INFLATED!**

Back to Example #1

- ANOVA (Analysis of Variance) is the appropriate procedure
- Some background of ANOVA
 - ANOVA was developed by R.A. Fisher in the early 1920
 - Initially applied to agricultural experiments
 - Used extensively today for industrial experiments
 - Probably the most useful technique in the field of statistical inference
- For this lecture, we will discuss the single-factor ANOVA (analysis of variance) for the fixed effect model
 - Fixed: it means that we only conclude about the factor levels considered in the analysis
 - Random: will be discussed later ...

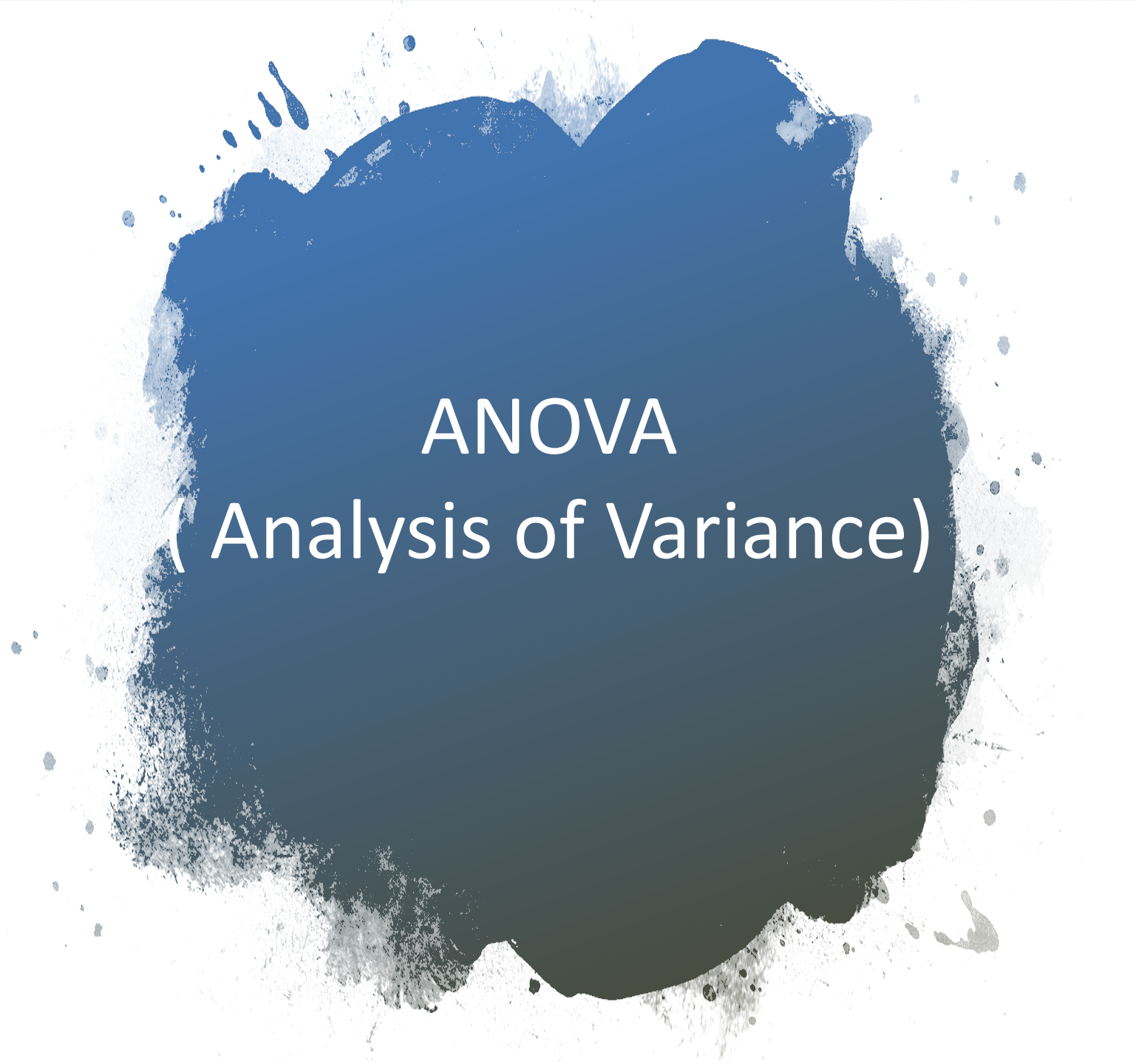
One more example: - single-factor experiment

one factor: percent of cotton
number of levels: 5

Tensile Strength Experiment

Investigate the tensile strength of a new synthetic fiber. The factor is the weight percent of cotton used in the blend of the materials for the fiber and it has five levels.

percent of cotton	tensile strength					total	average
	1	2	3	4	5		
15	7	7	11	15	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8



ANOVA (Analysis of Variance)

Suppose we have a treatments of a single factor, the **data layout** for single-factor experiments appears as follows:

treatment	observations				totals	averages
1	y_{11}	y_{12}	$\cdots$	y_{1n}	$y_{1.}$	$\bar{y}_{1.}$
2	y_{21}	y_{22}	$\cdots$	y_{2n}	$y_{2.}$	$\bar{y}_{2.}$
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$	$\vdots$
a	y_{a1}	y_{a2}	$\cdots$	y_{an}	$y_{a.}$	$\bar{y}_{a.}$

- y_{ij} represents the j_{th} observation taken under treatment i
- n observations under the i_{th} treatment
- $N = a \cdot n$ (total runs)
- All runs in random order
- A completely randomized design (CRD)

- Objective
 - To test hypotheses about the equality of a treatment means
- The name "Analysis of Variance" stems from a partition of the total variability in the response variable into components that are consistent w/ a model for the experiment
- First of all, let's look at statistical models for the single-factor ANOVA

Two Models

- Mean Model

$$y_{ij} = \mu_i + \varepsilon_{ij} \quad \left\{ \begin{array}{l} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{array} \right.$$

where

y_{ij} : the j th observation for treatment i

μ_i : the i th level mean

$\varepsilon_{ij} \sim N(0, \sigma^2)$

- Example:

- Salary of a student = college (that the student belong to) average salary + Random Error

Two Models – cont'd

- Effects Model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

Where

y_{ij} : the j th observation for treatment i

μ : the grand mean; $\mu_i = \mu + \tau_i$

τ_i : i_{th} treatment effect

$\varepsilon_{ij} \sim N(0, \sigma^2)$ – error

Constraints: $\sum_{i=1}^a \tau_i = 0.$

- Example:

– Salary of a student = RIT student average salary + college adjustment + Random Error

More on Effects Model

- Effects Model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{cases}$$

τ_i : i_{th} treatment effect $\leftarrow$ Fixed constant

Constraints: $\sum_{i=1}^a \tau_i = 0$.

So:

$$\sum_{i=1}^a (\mu_i - \mu) = 0, \quad \text{then} \quad \sum_{i=1}^a \mu_i - a\mu = 0$$

So that:
$$\mu = \frac{1}{a} \sum_{i=1}^a \mu_i$$

Two Models – cont'd

- Effects Model

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \quad \left\{ \begin{array}{l} i = 1, 2, \dots, a \\ j = 1, 2, \dots, n_i \end{array} \right.$$

- Mean Model

$$y_{ij} = \mu_i + \varepsilon_{ij}$$

- Both the mean model and the effects model are linear statistical models, and equivalent
- They are called the one-way or single-factor Analysis of Variance (ANOVA) models
 - Only one factor is investigated
- The **effects model** is more widely encountered in the experimental design literature

Basic Hypothesis

- If the factor has **no effect** on the response, then $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_a$ or $\tau_1 = \tau_2 = \tau_3 = \dots = \tau_a = 0$
- **The basic hypothesis**

$$H_0: \tau_1 = \tau_2 = \tau_3 = \dots = \tau_a = 0$$

$$H_a: \tau_i \neq 0 \text{ for at least one } i$$

or

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_a$$

$$H_a: \mu_i \neq \mu_j \text{ for at least one pair } (i,j)$$

❖ **ANOVA** is the appropriate procedure

- $y_{i.} = \sum_{j=1}^{n_i} y_{ij} \rightarrow \bar{y}_{i.} = y_{i.}/n_i$ (treatment sample mean, or row mean)
- $y_{..} = \sum \sum y_{ij} \rightarrow \bar{y}_{..} = y_{..}/N$ (grand sample mean)

Some Notations

One-way ANOVA

- **Basic idea**: to partition total variability of response into its component parts
- For simplicity, let's look at the **balanced case**: $n_i = n$
- **Total variability** in the data is measured by **the total sum of squares (SS_T)**

$$SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$$

- **Decomposition of y_{ij}**

$$y_{ij} = \bar{y}_{..} + (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$$

One-way ANOVA

- Estimates of parameters

$$\hat{\mu} = \bar{y}_{..}$$

$$\hat{\tau}_i = (\bar{y}_{i.} - \bar{y}_{..})$$

$$\hat{\epsilon}_{ij} = y_{ij} - \bar{y}_{i.} \quad (\text{residual})$$

$$\text{So } y_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\epsilon}_{ij}.$$

- It can be verified that

$$\sum_{i=1}^a n_i \hat{\tau}_i = 0; \quad \sum_{j=1}^{n_i} \hat{\epsilon}_{ij} = 0 \text{ for all } i.$$

One-Way ANOVA

- Recall:

$$y_{ij} - \bar{y}_{..} = \hat{\tau}_i + \hat{\epsilon}_{ij} = (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$$

- The **total sum of squares** SS_T can be written as:

$$\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = \sum_{i=1}^a \sum_{j=1}^n [(\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})]^2$$

- Expand the square** on the **right side**, we have

$$\begin{aligned} \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 &= n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 \\ &\quad + 2 \sum_{i=1}^a \sum_{j=1}^n (\bar{y}_{i.} - \bar{y}_{..})(y_{ij} - \bar{y}_{i.}) \end{aligned}$$

- The **cross term** is **zero**

One-Way ANOVA

Therefore, we have the basic ANOVA partitioning:

$$\begin{aligned}\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 &= \sum_{i=1}^a \sum_{j=1}^n [(\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})]^2 \\ &= n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2\end{aligned}$$

$SS_T = SS_{Treatments} + SS_E$

One-Way ANOVA

- The basic ANOVA partitioning:

$$SS_T = SS_{\text{Treatments}} + SS_E$$

Where

- SS_T : the total variability in the data
 - Measures how different are the y_{ij} 's from the grand average $\bar{y}_{..}$
- $SS_{\text{Treatments}}$: the sum of squares due to treatment (i.e., between treatments)
 - Measures how different are the treatment averages $\bar{y}_{i.}$ from the grand average $\bar{y}_{..}$
- SS_E : the sum of squares due to errors (i.e., within treatments)
 - Measures how different Y_{ij} 's from the the treatment average within a treatment $\bar{y}_{i.}$

Derive the test statistics

- Look at $SS_{\text{Treatments}} = \sum n_i \hat{\tau}_i^2 = \sum n_i (\bar{y}_{i.} - \bar{y}_{..})^2$
- A large value of $SS_{\text{Treatments}}$ reflects large difference in treatment means μ_i 's
- A small value of $SS_{\text{Treatments}}$ likely indicates no difference in treatment means μ_i 's
- If small then accept (not reject) H_0 ;
- If large then reject H_0 ;
 - But how small is small? how large is large?

Recall: Formal statistical hypotheses are:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_a$$

H_1 : At least one mean is different

Derive the test statistics – cont'd

- While **sums of squares cannot** be directly **compared** to test the hypothesis of equal means
- **Standardize to account for inherent variability**
- A **mean square** is a **sum of squares** divided by its **degrees of freedom**
 - “average” variability ($\frac{SS}{df}$)
- **Mean squares can be compared**

$$df_{Total} = df_{Treatments} + df_{Error}$$

$$an - 1 = a - 1 + a(n - 1)$$

$$MS_{Treatments} = \frac{SS_{Treatments}}{a - 1}, MS_E = \frac{SS_E}{a(n - 1)}$$

- Can show that the **expected values of MS** are

$$E(MS_E) = \sigma^2$$

$$E(MS_{Treatment}) = \sigma^2 + \sum n_i \tau_i^2 / (a - 1)$$

- If the **treatment means** are **equal**, the **treatment and error mean squares** will be (theoretically) **equal**.
- If **treatment means** differ, the **treatment mean square** will be **larger** than the **error mean square**

- Since, we have

$$E(MS_E) = \sigma^2$$

$$E(MS_{\text{Treatment}}) = \sigma^2 + \sum n_i \tau_i^2 / (a - 1)$$

- Hence, MS_E is an unbiased estimator of σ^2 , that is $\hat{\sigma}^2 = MS_E$.
 - Under H_0 , $MS_{\text{Treatments}}$ is also an unbiased estimator for σ^2

One-Way ANOVA

- Test Statistics

$$F_0 = \frac{SS_{\text{Treatments}} / (a - 1)}{SS_E / (N - a)} = \frac{MS_{\text{Treatments}}}{MS_E}$$

- Under H_0 , we know $E(MS_E) = E(MS_{\text{TREATMENTS}})$
- Hence, under H_0 , F_0 is expected to close to 1;
Large F_0 implied deviation from H_0
- What is the sampling distribution of F_0 under H_0 ?

Sampling Distribution of F_0 under H_0

- Under H_0 , $SS_{\text{Treatment}}/\sigma^2 = \sum_i n_i (\bar{y}_{i.} - \bar{y}_{..})^2 / \sigma^2 \sim \chi_{a-1}^2$
- Based on the model: $y_{ij} \sim N(\mu + \tau_i, \sigma^2)$ implies
 $\bar{y}_{i.} \sim N(\mu + \tau_i, \sigma^2/n_i)$ and $\sum_j (y_{ij} - \bar{y}_{i.})^2 / \sigma^2 \sim \chi_{n_i-1}^2$

therefore

$$SS_E / \sigma^2 = \sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2 / \sigma^2 \sim \chi_{N-a}^2$$

- $SS_{\text{Treatments}}$ and SS_E are independent
- by the definition of F distribution

$$F_0 = \frac{SS_{\text{Treatment}}/\sigma^2(a-1)}{SS_E/\sigma^2(N-a)} = \frac{\chi_{a-1}^2/(a-1)}{\chi_{N-a}^2/(N-a)} \sim F_{a-1, N-a}$$

- Under H_0 , F_0 follows a **F-distribution** with **degrees of freedom $a-1, N-a$**

One-way ANOVA Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Between	$SS_{\text{Treatment}}$	$a - 1$	$MS_{\text{Treatment}}$	F_0
Within	SS_E	$N - a$	MS_E	
Total	SS_T	$N - 1$		

- The **reference distribution** for F_0 is the $F_{a-1, a(n-1)}$ distribution
- Reject** the null hypothesis (equal treatment means) if

$$F_0 > F_{\alpha, a-1, a(n-1)}$$

- A p-value can also be computed; reject H_0 if **p-value** $< \alpha$

Some more computing formulas

- If **balanced** $n_i=n$; $N=a*n$

$$SS_T = \sum \sum y_{ij}^2 - y_{..}^2/N; \quad SS_{\text{Treatment}} = \frac{1}{n} \sum y_{i.}^2 - y_{..}^2/N$$
$$SS_E = SS_T - SS_{\text{Treatment}}$$

- If **unbalanced**: $N = \sum_{i=1}^a n_i$

$$SS_T = \sum \sum y_{ij}^2 - y_{..}^2/N; \quad SS_{\text{Treatment}} = \sum \frac{y_{i.}^2}{n_i} - y_{..}^2/N$$
$$SS_E = SS_T - SS_{\text{Treatment}}$$

Example

Twelve lambs are randomly assigned to three different diets. The weight gain (in two weeks) is recorded. Is there a difference between the diets?

Diet 1	8	16	9		
Diet 2	9	16	21	11	18
Diet 3	15	10	17	6	

- $N = 12$, $\sum \sum y_{ij} = 156$ and $\bar{y}_{..} = 156/12 = 13$.
- $n_1 = 3$, $y_{1.} = 33$, $\bar{y}_{1.} = 11$; $n_2 = 5$, $y_{2.} = 75$, $\bar{y}_{2.} = 15$; $n_3 = 4$, $y_{3.} = 48$ and $\bar{y}_{3.} = 12$.
- $\hat{\tau}_1 = \bar{y}_{1.} - \bar{y}_{..} = 11 - 13 = -2$; Similarly, $\hat{\tau}_2 = 15 - 13 = 2$ and $\hat{\tau}_3 = 12 - 13 = -1$.
- $SS_T = \sum_i \sum_j (y_{ij} - \bar{y}_{..})^2 = 246$.
- $SS_{\text{Treatment}} = 3 * (-2)^2 + 5 * (2)^2 + 4 * (-1)^2 = 36$.
- $SS_E = 246 - 36 = 210$

ANOVA Table

- $MS_E = \hat{\sigma}^2 = 210/(12 - 3) = 23.33$
- $F_0 = (36/2)/(210/9) = 0.77$; P-value > 0.25
- Fail to reject $H_0 : \tau_1 = \tau_2 = \tau_3 = 0$.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F0	P-value
Between	36	2	18	0.77	>0.25
Within	210	9	23.33		
Total	246				

Conclusion: there is **NO** significant difference between diets.

Design of Experiments

Worksheet 5 - One Way Analysis of Variance

The sales manager of ABC Software wants to see if dress code will have an effect on sales. Salespersons were randomly selected to wear business suits, casual outfits, or jeans when visiting prospective clients. The following sales (in thousands of dollars) for a four-week period have been recorded for each sales person:

Business Suits	Casual Outfits	Jeans
26	19	22
37	24	33
41	31	34
35	28	19
29	23	25
33	25	29
40	24	31
Mean =	Mean =	Mean =
Std Dev =	Std Dev =	Std Dev =

“Grand” Mean =
“Grand” Std Dev =

- A. Put each sample into a separate list in your calculator.
Then create a fourth list with the combined data.
- B. Compute the means and standard deviations that are needed above.
- C. Compute SSTotal as follows:
From the combined data, create a list of $(Y_{ij} - \bar{Y}_{..})$'s
Compute $\text{sum}(\text{list}^2)$ for this column.
SSTotal = _____
- D. Compute SSTreatment using the definition and the means you computed above. SSTreatment = _____
- E. Compute SSError as follows
From EACH separate sample, create a new list of $(Y_{ij} - \bar{Y}_{i.})$'s and compute $\text{sum}(\text{list}^2)$ for this column.
Add your three results together
SSError = _____

Design of Experiments

Worksheet 6 - One Way Analysis of Variance – Continued

Back to the Software Sales...

A. Verify that $SS_{\text{Total}} = (N - 1)s_Y^2$

B. Verify that $SS_{\text{Error}} = \sum_{i=1}^a (n - 1)s_i^2$

C. We often summarize data in an Analysis of Variance (ANOVA) table.

Source	SS	DF	MS	F	p-value
Treatment	SS_{Tr}	$a - 1$	$SS_{\text{Tr}}/DF_{\text{Tr}}$	$MS_{\text{Tr}}/MS_{\text{E}}$	$P(F > F_{\text{obs}})$
Error	SS_{E}	$N - a$	$SS_{\text{E}}/DF_{\text{E}}$		
Total	SS_{To}	$N - 1$	$SS_{\text{To}}/DF_{\text{To}}$		

NOTE: Can compute $P(F > F_{\text{obs}})$ with $\text{Fcdf}(F_{\text{obs}}, \infty, DF_{\text{Tr}}, DF_{\text{E}})$

Fill in the values below for the software sales data.

Source	SS	DF	MS	F	p-value
Treatment					
Error					
Total					

D. What hypotheses are you testing? Write them first using words (in context), then write using symbols.

E. What conclusion can you make from the data?

Design of Experiments Worksheet 7 – Formula Derivation

A. $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, i = 1, \dots, a, j = 1, \dots, n, \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$. Show that

$$Y_{ij} \sim N(\mu + \tau_i, \sigma_\varepsilon^2)$$

$$\bar{Y}_{i.} \sim N(\mu + \tau_i, \frac{\sigma_\varepsilon^2}{n})$$

$$\bar{Y}_{..} \sim N(\mu, \frac{\sigma_\varepsilon^2}{an})$$

Homework #2

B. Recall that $s_Y^2 = \frac{\sum_i \sum_j (Y_{ij} - \bar{Y}_{..})^2}{N - 1}$ and $s_i^2 = \frac{\sum_j (Y_{ij} - \bar{Y}_{i.})^2}{n - 1}$

Based on the results from part A, give two unbiased estimators of σ_ε^2

C. We know that $SS_{Total} = (N - 1)s_Y^2$ and $MS_{Total} = \frac{SS_{Total}}{N - 1}$

$$SS_{Error} = \sum_{i=1}^a (n - 1)s_i^2 \text{ and } MS_{Error} = \frac{SS_{Error}}{N - a}$$

Show that $E(MS_{Total}) = \sigma_\varepsilon^2$ and $E(MS_{Error}) = \sigma_\varepsilon^2$

Summary

- Today's lecture, we discussed
 - One-factor experiment
 - Completely randomized design
 - One-way ANOVA
 - Computing example
 - Manual computation
 - Worksheet 5+6
 - Can be downloaded
 - Will go through them next lecture
 - Homework 2 is assigned